2.23

a

$P(1N) = 0.95^1$

$P(7N) = 0.95^7 = 0.698337$

b

September has 30 days.

$P(Sept = 2) = \binom{30}{2}(0.05)^2(0.95)^28 = 0.2586367$

c

$10N = \{10 \text{ Days of no accidents}\}$

Therefore at least 1 accident in 10 days is $10N^c$

$P(10N^c|4N) = 1 - P(6N)$ because we know that the accident must occur within the last 6 days

$\rightarrow P(10N^c|4N) = 1 - (0.95^6) = 0.2469$

$P(10N^c \cap 4N) = P(10N^c|4N) * P(4N) = 0.2469 * 0.95^4 = 0.215769$

2.28

a

Hypergeometric distribution, $X\ Hypergeom(52, 4, 13)$

b

Binomial distribution, $X\ bin(50, 1 - \frac{48!39!}{35!52!}) \rightarrow X\ bin(50, \frac{14498}{20825})$

c

Binomial distribution, $X\ bin(50, \frac{12!39!}{51!} \rightarrow X\ bin(50, \frac{1}{8255176274800})$

d

Hypergeometric distribution, $X\ Hypergeom(52, 13, 13)$

2.47

$P(2F) = \frac{55}{80}\frac{54}{79} = \frac{297}{632}$

2.66

We can treat this problem as a binomial distribution with $X\ bin(n, \frac{1}{6})$ and we are solving for an n with the highest resulting probability.

$P(X = 2) = \binom{n}{2}(\frac{1}{6})^2(\frac{5}{6})^{n-2}$ for $n = 2, 3, ..., 40$

I will use the python code below to solve for n:

```
from math import comb
bestProb = 0
bestRolls = 0
for n in range(2,40):
  prob=(comb(n,2) * ((1/6)**2) *((5/6)**(n-2)))
  if prob > bestProb:
    bestProb = prob
    bestRolls = n
print(f"Optimal number of rolls is {bestRolls} resulting in a probability of {bestProb}")
```

```
Optimal number of rolls is 11 resulting in a probability of 0.2960935686313839
```

Therefore the optimal number of rolls is actually 11 resulting in a probability of 0.296

2.71

a

Let A_n denote the set where the nth flip is tails and $A_1 A_2 ... A_n$ represent a n consecutive tails flips.

Let $P(B)$ be the probability the coin is biased.

The question is solving for $P(B|A_1), P(B|A_1A_2), P(B|A_1...A_n)$

$P(B|A_1) = \frac{P(A_1|B)P(B)}{P(A_1)}$

From the problem (and exercise 2.38) we have the following values:

- $P(A_i) = \frac{51}{100}$
- $P(B) = \frac{1}{10}$
- $P(A_1|B) = \frac{3}{5}$

Plug those in to the above equation:

$$P(B|A_1) = \frac{P(A_1|B)P(B)}{P(A_1)} = \frac{3}{5}\frac{1}{10}\frac{100}{51} = \frac{2}{17}$$

For $P(B|A_1A_2)$:

$$P(A_1A_2|B) = \frac{3}{5}^2 = \frac{9}{25}$$

$$P(A_1A_2) = \frac{1}{2}^2\frac{9}{10} + \frac{3}{5}^2\frac{1}{10} = \frac{261}{1000}$$

$$P(B|A_1A_2) = \frac{P(A_1A_2|B)P(B)}{P(A_1A_2)} = \frac{9}{25}\frac{1}{10}\frac{1000}{261} = \frac{4}{29}$$

For $P(B|A_1...A_n)$:

$$P(A_1...A_n|B) = \frac{3}{5}^n$$

$$P(A_1...A_n) = \frac{1}{2}^n\frac{9}{10} + \frac{3}{5}^n\frac{1}{10}$$

$$P(B|A_1...A_n) = \frac{P(A_1...A_n|B)P(B)}{P(A_1...A_n)} = \frac{\frac{3}{5}^n\frac{1}{10}}{\frac{1}{2}^n\frac{9}{10}+\frac{3}{5}^n\frac{1}{10}} = \frac{\frac{3}{5}^n}{9\frac{1}{2}^n+\frac{3}{5}^n}$$

b

This question is for what n is $P(B|A_1...A_n) \geq 0.9$?

Below is the python code to help figure this out:

```
n=1
p=2/17

while(p < 0.9):
  n+=1
  p = ((3/5)** n) / (9*((1/2)**n) + (3/5)**n)
print(f"n = {n}: p = {p}")
```

```
n = 25: p = 0.9137899792797586
```

Thus after 25 straight tails we can say that with 90% probability that the coin is biased.

c

$P(A_{n+1}|A_1...A_n) = P(A_{n+1}|B)P(B|A_1...A_n) + P(A_{n+1}|F)P(F|A_1...A_n)$

$\rightarrow P(A_{n+1}|A_1...A_n) = P(A_{n+1}|B)P(B|A_1...A_n) + P(A_{n+1}|B^c)P(B^c|A_1...A_n)$

$\rightarrow P(A_{n+1}|A_1...A_n) = \frac{3}{5}\frac{\frac{3}{5}^n}{9\frac{1}{2}^n+\frac{3}{5}^n} + \frac{1}{2}(1 - \frac{\frac{3}{5}^n}{9\frac{1}{2}^n+\frac{3}{5}^n})$

d

$\lim_{n\to\infty} P(A_{n+1}|A_1...A_n) = \frac{3}{5}$ because $\lim_{n\to\infty} P(B|A_1...A_n) = 1$.

2.81

$|\Omega| = 669^n$

$U = \{\text{Unique birthdays}\}$

$U^c = \{\text{At least 2 with the same birthday}\}$

$|U| = P_n^{669}$

$P(U^c) = 1 - \frac{|U|}{|\Omega|} = 1 - \frac{669!}{669^n(669-n)!}$

Now for what n is $P(U^c) \geq 0.9$?

Again using python:

```
from math import perm
n=0
p=0
while (p< 0.9):
  n+=1
  p = 1 - (perm(669,n) / (669**n))
print(f"n = {n}: p = {p}")
```

```
n = 56: p = 0.9063675893321339
```

Therefore the n resulting in $P(U^c) > 0.9$ is $n = 56$, in other words a group of martians of size 56 has a 0.9 probability of having at least two matching birthdays.